



Full Proposal

Perception of fundamental frequency perturbations across many languages

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Abstract

Variations in onset fundamental frequency (f_0) of vowels have been systematically found to be influenced by the laryngeal states of adjacent consonants — the onset f_0 of a vowel is higher following a voiceless compared to a voiced obstruent. This phenomenon, the consonant-related f_0 perturbation (CF0), has been considered to be a key source of tonogenesis. Previous research on phonetic explanations for tonal development has examined listeners' ability to detect small f_0 movements, the use of f_0 perturbations as cues for laryngeal contrasts, and the emergence of f_0 perturbations as primary cues when other consonantal cues diminish. Despite extensive acoustic and perceptual studies, the understanding of CF0 perception remains limited. This project aims to investigate the perception of minimal perceptible f_0 perturbations, both in magnitude and timing, over many languages, and to gain insights on short-duration dynamic pitch perception in speech. Additionally, we propose to explore the effects of language experience and musical competence on CF0 perception. In this proposal, we detail the plans for developing and deploying an online framework for large-scale CF0 perception data that can be made use of by speech laboratories and fieldwork researchers across the globe so as to enrich our knowledge of CF0 informed by many languages.

1 Introduction

Over the past 70 years, it has been widely observed that variation in the fundamental frequency (f_0) of a vowel is systematically related to the attributes of its neighbouring consonant across typologically diverse languages (House and Fairbanks, 1953, and subsequent work. See §1.1 for a detailed review). The most robustly attested pattern is that f_0 , especially at vowel onset, is higher following a voiceless obstruent than following a voiced one. In this article, we refer to such consonant-related f_0 perturbations as CF0, following Kingston (2007). Various in the literature, CF0 is also known as “co-intrinsic pitch” (Di Cristo and Hirst, 1986), “micro f_0 ” (Kohler, 1990), “pitch skip” (Hanson, 2009), “onset voicing effect” (Kirby and Ladd, 2015), and “vertical perturbation” (Xu and Xu, 2021), reflecting the view of CF0 being an intrinsic effect attributed to the structure of the general speech mechanism (Wang and Fillmore, 1961; Ladefoged, 1965) and a part of microprosody (Di Cristo and Hirst, 1986). This interpretation, however, does not preclude the possibility that the seemingly automatic CF0 can be controlled or suppressed (see §1.2).

The phenomenon of CF0 has been regarded as ‘a seed’ of sound change, specifically, in the development of contrastive tones (Haudricourt, 1954; Matisoff, 1973; Thurgood, 2002). Following Hombert et al. (1979) and perceptual adaptation models (see §1.3), we require evidence of the following sort to provide a comprehensive phonetic explanation for tonal development:

- (1) Small CF0 perturbations should be detectable by listeners, so that in certain circumstances they may lead to the development of tone (minimal perceptible f_0 perturbations).
- (2) The f_0 perturbations associated with differences in consonants can serve as one of the perceptual cues to the consonant contrast (perceptual relations among cues).
- (3) When the salience of certain consonantal cues, such as voicing or aspiration, diminishes, f_0 perturbations can emerge as the dominant cue (cue weighting flexibility).

In addition to the extensive acoustic studies on CF0 across many languages, there is experimental evidence that small f_0 perturbations can serve as cues to discriminate consonant (laryngeal) contrasts (e.g. Haggard et al., 1970; Massaro and Cohen, 1976; Gao and Kirby, 2024), which indirectly demonstrate that CF0 perturbations can be detected by listeners. But our understanding of the capacity to perceive CF0 remains limited.

This project proposes to leverage the platform, expertise, and network of *ManyLanguages* to conduct a series of cross-linguistic perception experiments on consonant-related f_0 perturbations. This proposal will concentrate on the **first study** in this series on the minimal perceptible onset f_0 perturbations, inspired by the perception experiments in Hombert (1975, 1979). An introduction of Hombert’s experimental design will follow in §1.4.1. Speech and hearing research with human participants was often found to be underpowered (Gaeta and Brydges, 2020). With *ManyLanguages*, large-scale data will be possible to enable adequately powered and robust findings with cross-linguistic perspectives. The primary aims of this project include:

1. To enhance our knowledge on dynamic **pitch perception in speech**. This involves estimating the perceptual pitch equivalents for various vowel onsets with stylised f_0 movements, replicating and extending the experiments in Hombert (1975) to a larger participant pool with diverse language backgrounds. The sensitivity of the human ear to frequency perturbations is an crucial aspect of auditory discrimination. It is important to understand the distribution of ability in CF0 contexts across the population and the parameters which affect it .
2. To investigate **minimal f_0 perturbation and duration** required for onset pitch differentiation. Speech pitch discrimination is considered much harder than non-speech pitch discrimination (Kuang, 2013). The just noticeable difference (JND) for lexical tone appears to be at least 9 Hz (Silverman, 2003) and languages typically require 20 to 30 Hz difference to maintain a phonological contrast (’t Hart, 1981). To what extent can f_0 perturbations be perceived?
3. To explore whether and how **language experience** and **musical competence** affect CF0 perception. Considering that speech perception may involve auditory, phonetic, and phonological processing (Werker and Logan, 1985), conducting perception experiments across many languages offers insights into potentially common auditory processing and language-specific processing mechanisms. A growing body of research suggests that musicians generally show enhanced pitch discrimination abilities (e.g. Bianchi et al., 2016). Does musical competence impact CF0 perception? If so, musically talented and trained individuals within a speech community could potentially lead tonal change.
4. To create an **online framework and experimental workflow** to collect large-scale auditory perceptual data. The online framework would facilitate web-based auditory experiments conducted remotely over the internet, potentially broaden participant reach, and lower the burden on research labs.

All of the above insights will contribute to our understanding of CF0 perception, which is potentially a step forward in unravelling the underpinnings of tonal development, or the lack thereof. The following subsections present a cross-linguistic review of previous research on CF0 and pitch perception. This review not only furnishes a thorough background for this project, but also informs and motivates the design of the experiments (§2.1) and the formulation of our hypotheses (§1.4.2).

1.1 Consonant-related f_0 perturbations (CF0)

The pioneering study, House and Fairbanks (1953), identified a higher **mean f_0** of the vowel after voiceless consonants than after voiced consonants, produced by American English speakers. Lehiste and Peterson (1961) similarly reported that the **peak f_0** of the vowel is higher after voiceless consonants. Most subsequent studies concentrated on f_0 at the beginning of the vowel (**onset f_0**) following different consonants, where the greatest perturbation effect manifests.

The widely observed consonant-related f_0 perturbations are, however, not uniform across varieties and studies. The observed CF0 variations, in terms of direction, magnitude, and temporal extent, can be traced back to multiple sources, including the phonological status of the consonants (Dmitrieva et al., 2015), the local and global prosodic context (Kohler, 1982; Xu and Xu, 2003), as well as the method of the CF0 measurement and analysis ranging from the size of smoothing window for f_0 extraction to the alignment, reference, and time-normalisation of f_0 contours (Xu and Xu, 2021).

1.1.1 Direction of CF0

Early studies on American English such as Lea (1973) and Hombert (1978) concluded with a pattern of “rise-fall dichotomy” at the consonant-vowel transition; that is, the f_0 contour at vowel onset rises following a voiced obstruent and falls following a voiceless obstruent. The rise-fall dichotomy was also observed in other studies

such as Gandour (1974) on Siamese and Hombert (1977) on Yoruba, and has often been interpreted as two CF0 effects — f_0 raising by voiceless obstruents and f_0 lowering by voiced obstruents (Gao and Kirby, 2024).

Some other studies of English such as Haggard et al. (1981) and Ohde (1984) supported the alternative “no-rise” view that the f_0 falls after all stop consonants irrespective of their voicing or aspiration, though the extent of falling varies: the post-stop-release fall of f_0 is steeper after voiceless stops than after voiced stops. Hanson (2009) advanced the discussion on the direction of CF0 in American English by establishing the f_0 contour following nasal consonants as the baseline for comparison. She identified the f_0 raising after voiceless obstruents but not the f_0 lowering after voiced obstruents. The no-rise phenomenon was also observed in Danish (Jeel, 1975), Italian and French (Kirby and Ladd, 2015), and so on.

Considering the lack of consensus among studies, Xu and Xu (2021) highlighted the significant impact f_0 extraction and processing methods can have on the results. The trimming and smoothing, for instance, exaggerated the contrast of f_0 following voiced and voiceless consonants (ibid.). They identified a pronounced but brief f_0 raising at vowel onset, regardless of the voicing of the preceding consonant, and suggested that when such initial f_0 jumps are preserved, the f_0 fall after both voiced and voiceless consonants (no-rise) tends to be observed. Despite the controversy regarding the CF0 direction after voiced obstruents, one critical finding is that f_0 is consistently raised following (phonologically) voiceless stops.

1.1.2 Magnitude of CF0

The reported magnitude of CF0 perturbations varies greatly among studies. Tables 1 and 2 summarise the magnitude of CF0 across 37 languages reported in the literature, quantified as the difference (Δf) between mean f_0 s following different types of plosives, distinguished by their voicing ($\Delta f(\text{C} - \text{C})$, voiceless – voiced) or aspiration ($\Delta f(\text{C} - \text{C}^h)$, voiceless unaspirated – voiceless aspirated). A positive Δf indicates a higher f_0 following the former type of stops, while a negative Δf indicates a higher f_0 following the latter. Given the typological rarity of languages with prevoiced aspirated stops, the difference between f_0 following voiced unaspirated and voiced aspirated stops is not presented in the table. Comparisons should be made with great caution, as these studies span over a few decades, during which recording devices and acoustic analysis techniques have drastically improved. Furthermore, many studies involved very small sample sizes in terms of the number of speakers and items.

As shown in Tables 1 and 2, the f_0 differences between post-voiceless and post-voiced vowel onsets are consistently positive, ranging from as low as 3 Hz to over 50 Hz. Even among the extensive research on English, these differences at vowel onset vary from 3 Hz to 38 Hz. The f_0 differences between post-voiceless unaspirated and post-voiceless aspirated vowel onsets can, nevertheless, be either positive or negative, and this variation may be language-specific. The signs of these f_0 differences, though, are sometimes inconsistent even among studies within the same language (e.g. Cantonese, Mandarin, and Thai). The magnitude also varies within and across languages. Voiceless unaspirated stop consonants tend to exhibit variable CF0 effects across many languages, potentially due to their differing phonological statuses (Dmitrieva et al., 2015).

Many CF0 differences are larger than 20 Hz, even in tone languages such as Yoruba, Punjabi, Xhosa, Burmese, and Shanghai Wu. This is a bit surprising since the typical pitch difference between tones is about 20 Hz (Kuang, 2013, Table 1) in different tone languages with varied tone levels. There is, therefore, **no sufficient** evidence to support the previous claim that CF0 differences tend to be relatively large in non-tonal languages and very small in tone languages (Hombert, 1978).

1.1.3 Temporal extent of CF0

Non-contrastive CF0 perturbations are typically localised or maximised at the vowel onset. In Italian and French, for example, the CF0 differences at vowel onset (see Table 1) quickly diminish and are no longer significant by vowel **midpoint** (Kirby and Ladd, 2015). Similarly, Hanson (2009) mentioned the absence of systematic f_0 differences at mid-vowel for English vowels following voiced and voiceless obstruents. As Dmitrieva et al. (2015) reported, such f_0 differences ceased being significant at approximately 75 ms into the vowel in English, and at 53 ms in Spanish.

Many studies observed a tendency for CF0 differences to converge earlier in the vowel in tone languages than non-tonal languages¹. For example, in Thai the perturbation effect on f_0 lasts for about 30–50 ms into a vowel (Gandour, 1974), in Yoruba about 40–60 ms into a vowel in low or mid tone (Hombert, 1977), in Cantonese about 20–40 ms into a vowel in high level or rising tone (Francis et al., 2006). Although Lai et al. (2009) found

¹For ease of exposition, we are categorising Danish, Swedish, Japanese, and Swahili as non-tonal languages, however, we are aware that these languages are reported to have register, stød, pitch accents (especially, the Tokyo variety of Japanese), and word accents respectively.

Non-tonal languages

Language Variety	Source	Measurement	$\Delta f(\text{C}_\circ - \text{C})$	$\Delta f(\text{C}_\circ - \text{C}^h)$
Afrikaans	Coetzee et al. (2018a, FIG 3)	Onset	44 Hz	
Bulgarian	Ting et al. (2023, TABLE 5)	Onset	2.72 st	
Catalan	Pricop and Chodroff (2024, FIG 1)	First 10% point	4 Hz	
Croatian	Ting et al. (2023, TABLE 5)	Onset	1.16 st	
Czech	Ting et al. (2023, TABLE 5)	Onset	2.20 st	
Danish	Jeel (1975, TABLE I)	Onset	12.88 Hz	
Dutch	van Alphen and Smits (2004, FIG 9)	Onset	16 Hz	
English (American)	House and Fairbanks (1953, TABLE III)	Average	6 Hz	
English (American)	Lehiste and Peterson (1961, TABLE IV)	Peak	12 Hz	
English (American)	Lea (1973, FIG 7 & 8)	Onset	30 Hz	
English (American)	Hombert (1978, FIG 1)	Onset	17 Hz	
English (Southern British)	Haggard et al. (1981, FIG 1)	Onset	14 Hz	
English (American)	Ohde (1984, FIG 2)	Onset	32 Hz	
English (Californian)	Jun (1996, FIG 1)	Onset	27 Hz	
English (American)	Hanson (2009, FIG 7)	Onset	38 Hz	
English (American)	Dmitrieva et al. (2015, TABLE 2 & 3)	Onset	-0.44 st (within [+voice])	
English (American)	Xu and Xu (2021, TABLE II & FIG 11)	Onset	2.03 st	
English (American)	Ting et al. (2023, TABLE 5)	Onset	3 Hz (untrimmed)	
English (American)	Ting et al. (2023, TABLE 5)	Onset	15 Hz (trimmed)	
English (American)	Ting et al. (2023, TABLE 5)	Onset	2.07 st	
French (Paris)	Jun (1996, FIG 3)	Onset	37 Hz	
French	Kirby and Ladd (2015, FIG 4)	Onset	12 Hz	
French	Ting et al. (2023, TABLE 5)	Onset	3.5 st	
German	Kohler (1982, TABLE I, II & V)	Onset	17 Hz	
German	Ting et al. (2023, TABLE 5)	Onset	0.91 st	
Hindi	Kagaya and Hirose (1975, TABLE II)	Onset	34 Hz	10 Hz
Hindi	Shimizu (1994, TABLE 1)	Onset	14.7 Hz	8.1 Hz
Italian	Kirby and Ladd (2015, FIG 4)	Onset	12 Hz	
Japanese	Shimizu (1994, TABLE 1)	Onset	34.7 Hz	
Korean (Seoul)	Han and Weitzman (1970, TABLE II)	Onset		-42 Hz (lenis)
Korean (Seoul)	Shimizu (1994, TABLE 1)	Onset		-14 Hz (lenis)
Korean (Seoul)	Jun (1996, FIG 4)	Onset		-58 Hz (lenis)
Korean (Chonnam)	Choi (2002, FIG 4)	Onset		-36 Hz (lenis)
Korean (Seoul)	Choi (2002, FIG 4)	Onset		-47 Hz (lenis)
Korean (Seoul)	Cho et al. (2002, FIG 5)	Onset & mid		-18 Hz (lenis)
Korean (Cheju)	Cho et al. (2002, FIG 5)	Onset & mid		-30 Hz (lenis)
Korean	Ting et al. (2023, TABLE 5)	Onset		-0.81 st (tense)
Madurese	Misnadin et al. (2015, FIG 2)	Onset	10.5 Hz	-0.8 Hz
Polish	Ting et al. (2023, TABLE 5)	Onset	2.92 st	
Portuguese (Brazilian)	Ting et al. (2023, TABLE 5)	Onset	2.17 st	
Russian	Ting et al. (2023, TABLE 5)	Onset	1.66 st	
Spanish	Dmitrieva et al. (2015, TABLE 2)	Onset	1.24 st	
Spanish (Latin American)	Ting et al. (2023, TABLE 5)	Onset	1.03 st	
Swahili	Ting et al. (2023, TABLE 5)	Onset		-1.35 st
Swedish	Löfqvist (1975, TABLE II & IV)	Peak	14 Hz	
Swedish	Ting et al. (2023, TABLE 5)	Onset	1.03 st	
Turkish	Ting et al. (2023, TABLE 5)	Onset	1.45 st	
Ukrainian	Ting et al. (2023, TABLE 5)	Onset	3.9 st	

Table 1: The magnitude of plosive-related f_0 perturbations in non-tonal languages by voicing and aspiration attributes in chronological order. Results were **aggregated** across various conditions in each paper and some f_0 values were estimated from the figures in these papers. Note that the papers on English mostly consider short-lag voiceless plosives (/p, t, k/) as phonologically voiced (/b, d, g/). Similarly, the voicing contrast (word initially) is dependent on the presence or absence of aspiration in German and Danish. Findings from Ting et al. (2023) are based on CF0 estimates of obstruents derived from the Generalised Additive Mixed Models.

Tone languages

Language Variety	Source	Measurement	$\Delta f(C_c - C)$	$\Delta f(C_c - C^h)$
Burmese	Shimizu (1994, TABLE 1)	Onset	20.7 Hz	0.9 Hz
Cantonese	Zee (1980, TABLE I)	Onset		-9 Hz
Cantonese	Francis et al. (2006, FIG 1)	Onset		11.5 Hz
Cantonese	Ting et al. (2023, TABLE 5)	Onset		-0.34 st
Hausa	Ting et al. (2023, TABLE 5)	Onset	2.06 st	
Khmer	Kirby (2018, FIG 2)	Onset	1.35 st	-0.85 st
Kurtöp	Hyslop (2009, FIG 2 & 3)	Onset	20 Hz	3 Hz
Mandarin	Shimizu (1994, TABLE 1)	Onset		7.1 Hz
Mandarin	Xu and Xu (2003, FIG 5)	Onset		30 Hz
Mandarin	Ting et al. (2023, TABLE 5)	Onset		-0.47 st
Thai	Gandour (1974, TABLE III)	Onset	20 Hz	12 Hz
				17.8 Hz (C_c^h)
Thai	Shimizu (1994, TABLE 1)	Onset	2.5 Hz	-19.1 Hz
Thai (Bangkok)	Kirby (2018, FIG 3)	Onset	0.77 st	-0.76 st
Thai	Ting et al. (2023, TABLE 5)	Onset	1.39 st ($C_c^h - C$)	
Pahari	Rashid et al. (2021, TABLE 1)	Onset	15 Hz	
Punjabi	Mirza (1990, FIG 1)	Onset	53 Hz	
Shanghai Wu	Chen (2011, FIG 4)	Onset	42 Hz	35 Hz
Taiwanese	Lai et al. (2009, TABLE 1)	Onset		-3.3 Hz
Vietnamese (Hanoi)	Kirby (2018, FIG 4)	Onset	1.03 st	-0.44 st
Vietnamese	Ting et al. (2023, TABLE 5)	Onset	1.27 st	
Xhosa	Jessen and Roux (2002, FIG 1)	Onset	23 Hz	
Yoruba	Hombert (1977, FIG 2)	Onset	37 Hz	

Table 2: The magnitude of plosive-related f_0 perturbations in tone languages by voicing and aspiration attributes in chronological order. Results were **aggregated** across various conditions in each paper and some f_0 values were estimated from the figures in these papers. Findings from Ting et al. (2023) are based on CF0 estimates of obstruents derived from the Generalised Additive Mixed Models.

the CF0 effect to continue for as long as 140 ms in Taiwanese, this duration in fact only represents up to 30% of a tonal contour, considering that some tones extend over 550 ms.

There are cases, however, where CF0 differences persist throughout a vowel, as observed in Afrikaans (Cotzee et al., 2018a, Fig. 3), Seoul Korean (Jun, 1996, p.103), and Kurtöp (Hyslop, 2009, Fig. 2 & 3). In Ting et al. (2023)’s model estimates based on corpus data, while the CF0 difference tapers off well before the vowel midpoint in some languages (e.g. Swedish, Swahili), it persists throughout the entire vowel in others (e.g. Bulgarian, Polish). In Korean, the trend of enhanced f_0 distinction been seen as evidence of phonologisation (Jun, 1996) and suggested as a stage in Seoul Korean tonogenesis (Silva, 2006; Kang, 2014). Similarly in Kurtöp, which is currently undergoing tonogenesis, the statistically significant CF0 difference is considered phonologised (Hyslop, 2009).

1.2 Explanations of the f_0 perturbations

The covariation between voicing and f_0 is understood to be governed by aerodynamic and biomechanical constraints and has been argued to arise from automatic effects of voicing production, including the stiffening and slackening of the vocal folds and the raising and lowering of larynx (Halle and Stevens, 1971; Honda, 2004). For instance, voicing during stop closure is enabled by lowering the larynx, thereby expanding the supraglottal cavity to allow vocal fold vibration to continue despite oral closure. Concurrently, relaxation of the cricothyroid muscle resets vocal fold tension, resulting in f_0 lowering (Honda, 2004). Additionally, the cricoid cartilage moves downward, aligning the posterior plate parallel to the curvature of the cervical spine. This rotation of the cricoid cartilage leads to the shortening of the vocal folds, also contributing to the lowering of f_0 .

On the other hand, this covariation appears to be determined by the obstruent’s phonological specification, regardless of how the voicing contrast is phonetically manifested, whether in aspirating languages or in “true-voicing” languages (Dmitrieva et al., 2015). In other words, f_0 is raised next to voiceless unaspirated stops in languages such as French but lowered next to such stops in languages such as English (Kingston, 2007). CF0 has

thus been alternatively posited to function as an auditory enhancement for the [voice] contrast under speaker control (Whalen et al., 1993; Kingston and Diehl, 1994). Prosodic contexts may also play a role in CF0 variations. In Korean and Japanese, for example, CF0 perturbations are enhanced in focus positions and compressed in pre- and post-focus positions for both narrow and contrastive focus (Chen et al., 2023). A unified or hybrid account has also been proposed in previous studies (e.g. Kingston and Diehl, 1994; Holt et al., 2001; Hoole and Honda, 2011), incorporating controlled factors such as enhancement alongside physiologically governed automaticity.

1.3 Perception research on f_0 perturbations

1.3.1 Differential sensitivity to pitch distance in speech

Early psychoacoustic research investigated the accuracy with which pitch can be perceived and particularly the just-noticeable difference (JND) for f_0 under various conditions. Using an analog formant synthesizer and pulse train generator, the pioneering work of Flanagan and Saslow (1958) reported a JND of 0.3–0.5 Hz for static synthetic vowels of 80 and 120 Hz. The relevance of such precise f_0 discrimination to speech perception, however, remains questionable (Klatt, 1973). Much higher JND values for f_0 in static conditions with resynthesised speech signals have also been reported; for instance, Isačenko and Schädlich (1970) obtained 5% (of 150 Hz).

It is established that relative or dynamic changes in f_0 are more important than absolute pitch levels in speech (Lieberman, 1965). A series of early studies has focused on the perceptual thresholds for f_0 glissando (unidirectional frequency modulation) in pure tones (e.g. Sergeant and Harris, 1962; Pollack, 1968; Nabelek and Hirsh, 1969; Schouten, 1985) and in synthetic vowels (e.g. Rossi, 1971; Klatt, 1973), though the findings show large discrepancies. The discrimination of f_0 changes in linear ramp f_0 contours can be an order of magnitude worse than in static f_0 contours. For example, Klatt (1973) reported a JND of 2 Hz when the f_0 contour of a 250-ms synthetic vowel /ε/ was a linear descending ramp (32 Hz/sec), as opposed to a JND of 0.3 Hz for a constant f_0 (of 120 Hz). Rossi (1971), instead, reported a glissando threshold of 19 Hz for a 200-ms natural vowel with an upward glide starting at an f_0 of 135 Hz. 't Hart (1976) approached the glissando threshold (G_t in semitones/s) by $G_t = 0.16/T^2$, where T is the tone duration. Below this threshold, an f_0 glissando is perceived with constant pitch, without a sensation of glissando. Rossi (1971, 1978) modelled the pitch perceived at the end of vowels with short-duration glissandos using the “2/3 rule”, which corresponds to a point between the second and the third thirds of the vowel. Other numerical models include the weighted time-average model (d'Alessandro et al., 1998).

The traditional JND for a synthetic vowel is comparable to, and even slightly smaller than, that for a pure tone of the same frequency and sound pressure level (Harris, 1952). This is often attributed to the presence of a harmonic structure in a synthetic vowel, which may provide additional information to the auditory system. However, JND values for resynthesised and natural speech materials appear to be much larger. Although results from previous studies cannot be directly compared due to differences in stimuli duration, frequency, and experimental paradigms, evidence suggests that speech is a significantly more challenging medium for the discrimination of dynamic pitch than non-speech (Turner et al., 2019). 't Hart (1981) measured the JND of the size of pitch movements in conditions representative of the dynamic situation in speech and suggested that only differences of at least 3 semitones play a part in communicative situations. Turner et al. (2019) reported a JND of 35 Hz for speech rises and 40 Hz for speech falls, in contrast to 17 Hz for non-speech rises and 25 Hz for non-speech falls. In Liu (2013), significantly lower JNDs were found for resynthesised Mandarin and English vowels than JNDs for nonspeech tonal glides, but only in level tones. For both native Mandarin and English speakers, the JNDs for dynamic tones ranged from 3 to 14 Hz, and the variations were associated with the size f_0 change and the position of f_0 judgments (i.e. onset or offset).

The perception of CF0 essentially involves the perception of short-duration f_0 glissandos in complex speech contexts. The present study builds on the JND literature and explores f_0 glissando scenarios relevant to CF0. A comparison of data from pure tones and speech-like stimuli investigates whether CF0 perception operates as a general auditory mechanism or is modulated by language experience. If language experience does play a role, data from many languages further explores its specific effects.

1.3.2 The necessity of studying CF0 perception across languages

Sound changes involve the changes of cue weighting in perceiving covarying acoustic features (Beddor, 2009; Harrington et al., 2019; Gao and Kirby, 2024; Schertz and Clare, 2020). Beddor (2009; 2012) exemplified this with VN vowel nasalization in American English, where the vowel is nasalized because of the coarticulatory effect: the presence of N is the coarticulatory source, while the nasalized vowel is the effect. When perceiving this VN sequence, listeners may regard the nasalized vowel and the nasal consonant as perceptually equivalent.

The underlying mechanism is that the articulation of the nasalized vowel and nasal consonant both involve an overlapping gesture – the lowering of velum. Over the passage of time, listeners rely more on the effect than the source, leading to sound change. In the case of vowel nasalization, Beddor (2009) also found empirical perceptual evidence: listeners were only sensitive to the duration of nasality but could not distinguish whether the nasal element was associated with the vowel or consonant. In other words, they were unable to tell apart the source from the effect. This perceptual evidence can be used to indirectly support that perceptual equivalence and cue-weighting can reasonably account for sound change.

Extending Beddor’s findings, many researchers have attempted to use the trading between source and effect to account for the CF0 effect in tonogenesis (Coetzee et al., 2018b; Gao and Kirby, 2024; Kuang and Cui, 2018; Yang et al., 2015): the VOT of the onset consonant and the onset pitch of the following vowel covary, with the former being the source and the latter being the effect. To make the voicing contrast of consonants, the more listeners rely on the vowel onset pitch, the less they rely on the VOT difference, which gives rise to tones. Similar to velum lowering in VN-nasalisation, the source-effect trading account of the CF0 effect also has an articulatory basis, though the exact mechanism remains debated, as CF0 involves more complex gestures (see §1.2) than nasalization (Gao and Kirby, 2024). Due to this complexity, it is not immediately clear how well the cue-weighting or source-effect trading account explains tonogenesis from an articulatory perspective.

From a macroscopic perspective on tonogenesis, if tones indeed originate from varying listener reliance on different cues, speakers of languages already possessing tones may be more attuned to the effect itself, namely pitch, and less concerned with the source, such as the onset voicing or aspiration that caused minor perturbations in f_0 . Alternatively, it is also possible that due to the presence of tones, there could be a cumulative effect between tone and CF0, potentially reducing tonal language speakers’ sensitivity to the consonant voicing caused f_0 perturbations. For listeners of non-tonal languages, similar to pre-tonogenesis speakers, there remains a possibility of transitioning from reliance on VOT to sensitivity towards f_0 changes. Since it is inconclusive whether tonal language speakers have an advantage in perceiving the perturbation compared to non-tonal speakers, it is crucial to collect more cross-linguistic perceptual data to investigate the potential differences.

1.3.3 The effects from tonal experience and musical ability

Language experience affects the extent to which speakers are sensitive to certain speech sounds, as they map these sounds to discrete categories shaped by prior language exposure, resulting in categorical perception (e.g. Liberman et al., 1957). Categorical perception begins in early infancy. During the first year after birth, infants’ ability to discriminate native phonemic contrasts gradually strengthens, while their ability to discern phonemic contrasts in non-native languages diminishes (Gervain and Mehler, 2010; Kuhl et al., 2006; Mattock and Burnham, 2006). The development of categorical perception occurs not only in segment perception (e.g., place of articulation: Liberman et al., 1957; manner of articulation: Fitch et al., 1980; vowel duration: Minagawa-Kawai et al., 2007) but also in lexical tone perception (Chen et al., 2020; Francis et al., 2003; Hallé et al., 2004; Peng et al., 2010; Wu et al., 2015; Zheng et al., 2012).

Equipped with native phonemic categories, speakers tend to assimilate novel L2 sounds to the categories in their L1, which can be seen as evidence for tonal categorical perception. The perceptual assimilation in second language acquisition, noted in Best’s Perceptual Assimilation Model (PAM, 1995), has also been extended to tonal research. It has been found that the perceptual assimilation of lexical tones occurs both phonetically and phonologically for speakers whose L1 is tonal. For example, in Chen et al. (2020), Mandarin, Northern Vietnamese, and Southern Vietnamese speakers, who were naïve to Thai, were asked to assimilate Thai tones into their native tonal categories. They found that Mandarin speakers rated the mid-level tone in Thai as similar to the only level tone, high-level tone, in Mandarin; Northern Vietnamese speakers assimilated both the high-rising tone (T45) and the falling-rising tone (T315) in Thai to the only rising tone in their native language; Southern Vietnamese speakers assimilated both the low-falling (T21) and mid-level tones (T33) to their low-level tone (T22). These results support the phonological assimilation assumption. Additionally, Chen et al. (2020) found that participants showed sensitivity to within-category phonetic differences, reflected in their longer response times to novel tones that were harder to categorise. In summary, tonal speakers’ perception of non-native pitch differences or changes is “pulled” towards their native tonal categories, but phonetic residuals remain.

For speakers without a tonal background, it is possible that they assimilate novel tones to their native intonation categories. However, such assimilation is fundamentally different from lexical tone categorical assimilation (Best, 2019) and is much weaker (Francis et al., 2008). Consequently, non-tonal speakers tend to perceive tones psychoacoustically, being less influenced by existing categories. Indeed, it has been suggested that while tonal speakers show decreased sensitivity to within-category pitch differences, non-tonal speakers are more sensitive to absolute pitch differences (Francis et al., 2003; Stageray and Downs, 1993; Tanner Jr and Rivette, 1964). Crucially, categorical perception of tones can develop in non-tonal speakers through musical experience, as musical

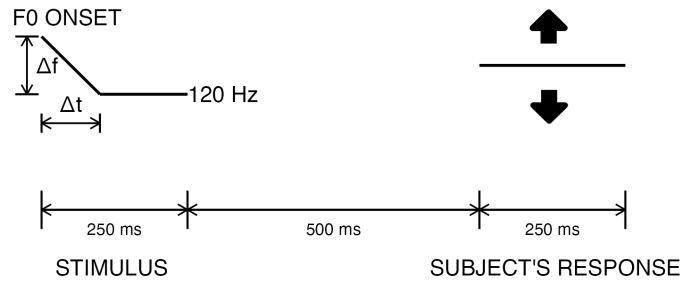


Figure 1: Schematic representation of the experimental paradigm of Hombert (1975, p.223 Part I).

notes are discrete (Howard et al., 1992; Zatorre and Halpern, 1979).

In addition to the listener’s tonal background, Music experience significantly enhances pitch-matching abilities, as evidenced by various behavioural and neuroscientific studies. Research indicates that musical training refines auditory perception, enabling musicians to discern and reproduce pitches with greater accuracy. Gaab and Schlaug (2003) investigated the effect of musicianship on pitch memory by comparing musicians and non-musicians matched for performance. The results demonstrated that musicians exhibited superior pitch memory, suggesting that musical training enhances auditory working memory capabilities. Moreover, Arndt et al. (2020), Proverbio et al. (2016) and Moritz (2023) show that musicians have smaller Just Noticeable Difference (JND) thresholds compared to non-musicians and finer more tuned perception of pitch differences, when they hear pure tones. Arndt et al. (2020) examined the pitch discrimination abilities of musicians and non-musicians using an oddball paradigm while recording EEG activity. The findings revealed that musicians had significantly smaller just-noticeable differences (JNDs) for pitch, indicating heightened sensitivity to frequency changes. Electrophysiological measures such as mismatch negativity (MMN) and P3a responses were more pronounced in musicians, reflecting enhanced neural processing of pitch deviations.

Previous studies have primarily focused on the comparisons between (non)tonal musicians and non-musicians. However, defining the boundary between musicians and non-musicians is difficult, particularly when we take cultural and educational differences into account (i.e., not everyone has received formal musical education of the same kind). Furthermore, even laypeople can be implicit learner of music through daily exposure, especially with modern technology (e.g. smart phones), leading to thousands of hours of musical experience by adulthood. Therefore, we will use music aptitude instead of musicianship as a measure of musical competence, employing the Musical Ear Test (Wallentin et al., 2010) to assess how sensitive participants are to melodic differences. This study incorporate the potential musical effects in the perception analysis and applies regression analysis to probe into the relationship between music aptitude and dynamic pitch-matching performance.

1.4 The current study

This study adapts the pitch-matching paradigm in Hombert (1975) to investigate the perceptual sensitivity to onset f_0 perturbations in incrementally speechlike stimuli, ranging from pure tones to resynthesised vowels and CV syllables. Data from many languages not only provide a more representative sample for robust auditory research compared to previous studies, but also allow for testing whether language experience modulates CF0 perception.

The majority of the previous studies on pitch perception employed a forced-choice paradigm in pitch discrimination tasks. However, the pitch-matching paradigm offers several advantages. With each response it outputs an estimation of the pitch perceived by a subject, and the adjustment procedure keeps subjects actively engaged and concentrated (d’Alessandro et al., 1998). Moreover, the data obtained from the **adjustment procedure** are less variable than forced-choice data, according to Wier et al. (1976).

1.4.1 The pitch-matching paradigm in Hombert (1975)

Figure 1 illustrates the pitch matching paradigm used in Hombert (1975) to investigate the perceptibility of CF0. Acoustic stimuli consist of instances of synthesised vowel [i] with stylised f_0 patterns, simulating the effects of obstruents on neighbouring vowels. The f_0 contour of each stimulus was composed of a slope followed by a level tone at 120 Hz. The onset frequency of the slope varied at 70, 100, 110, 130, 140, and 190 Hz (that is, $\Delta f = \pm 10$,

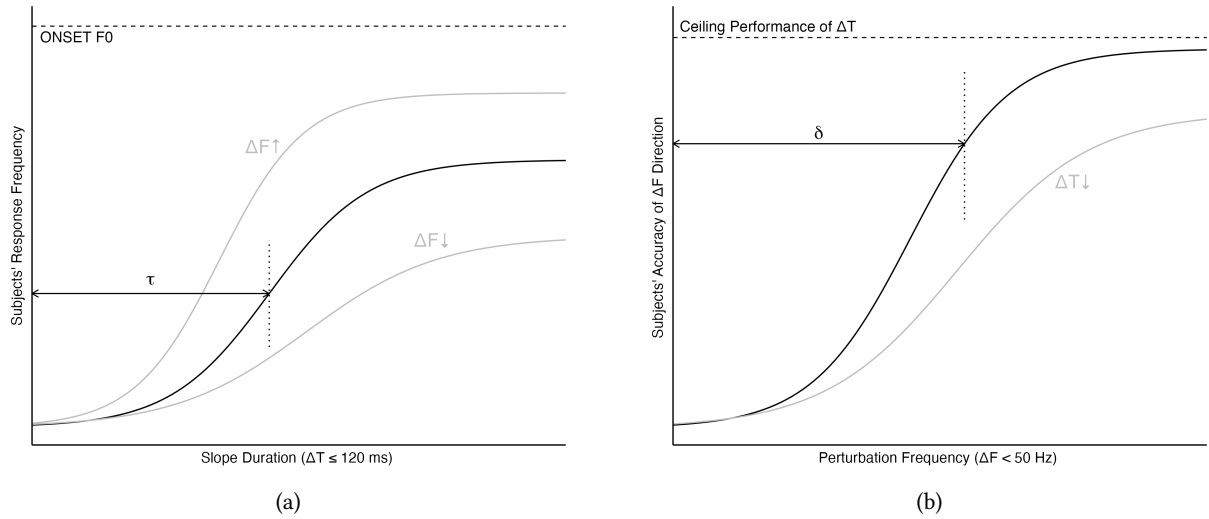


Figure 2: (a) Possible functions fitted to subjects' response frequency against slope duration. τ represents the slope duration threshold needed to perceive the onset f_0 perturbation. Grey curves signal how the function may change given different onset f_0 (ΔF). (b) Possible functions fitted to subjects' average accuracy in detecting the direction of ΔF in ΔT condition. δ represents the JND threshold at 75% of the ceiling performance. Grey curves signal how the function may change given different onset slope duration (ΔT).

± 20 , -50 and $+70$ Hz). The duration (Δt) of the slope was varied at 40, 60, 100, 150, or 250 ms. The overall duration of each stimulus was fixed at 250 ms. 500 ms after a stimulus was presented, participants were asked to match the pitch of the second vowel with a steady state f_0 to the pitch of the **beginning** of the stimulus vowel by adjusting a knob. The rate of stimulus presentation and the number of trials for a given presentation were controlled by the subject. Each of the stimuli was presented three times in a randomised order.

The original experiment in Hombert (1975), however, involved only ten subjects, all native speakers of American English. The findings are hardly generalisable given the very small sample size. It remains unclear whether and how long-term linguistic experience and musical competence influence the cognitive processing of pitch contours related to the 'phonetic universal' of CF0.

1.4.2 Hypotheses

This empirical study on the perception of CF0 across many languages is exploratory in nature. With the pitch-matching paradigm, some possible results are schematically illustrated in Figure 2. We have the following general hypotheses:

- Hypothesis 1 The longer the slope, the more accurate the pitch matching** A longer slope duration will lead to more accurate pitch matching by listeners, as observed in Hombert (1975). As in Figure 2a, a threshold of slope duration τ is expected, above which a separate perception of the onset f_0 perturbation and the level tone appears. As illustrated by the sigmoid-shaped psychometric function in Figure 2b, a JND threshold δ is also expected for a given slope duration, above which the proportion of correct judgments on the direction of pitch change is significantly above chance level. With data of perceived pitch, numerical models can be developed to simulate and predict what pitch is subjectively perceived at the beginning given an f_0 perturbation.
- Hypothesis 2 Linguistic versus non-linguistic stimuli** Comparing the perception of identical f_0 patterns presented in linguistic such as pure tones and non-linguistic contexts such as vowels and syllables allows us to test whether CF0 perception is acoustic or linguistic in nature, although we are primarily interested in the speech experiments. If acoustic in nature, perceptual effects should be maintained across stimuli conditions. If linguistic in nature, then we predict that pitch matching accuracy will be higher for non-linguistic stimuli than for linguistic stimuli, as the JND literature suggests that pitch perception in speech is significantly more difficult than in its nonspeech analogues (Turner et al., 2019).
- Hypothesis 3 Tonal speakers versus non-tonal speakers** For nonspeech stimuli, we predict that listeners from different language backgrounds will show identical responses, controlling for the musical training and talent. For speechlike stimuli, non-tonal speakers' performance will surpass that of tonal speakers. When

tone-language speakers perceive CF0 perturbation (usually much smaller than the between-category tonal differences), their responses may be influenced by higher-level categorical structure, thereby assimilating the CF0 contour to the base level tone. Previous research on the effect of linguistic background on pitch or tonal perception has yielded inconsistent results (Bent et al., 2006). Rather than categorising languages into a strict tonal dichotomy, it is worth considering the density of the tonal space or the size of tonal dispersion (e.g. a language with five level tones versus one with two tones). In line with the previous prediction, the perception of CF0 is expected to be inhibited in languages with large tone inventories. Prior to cross-linguistic comparison, perceptual responses of individual languages will be analysed and interpreted with respect to their typical CF0 acoustics and phonological categories.

Hypothesis 4 Musician versus non-musician For auditory experiments, it is hard to interpret the results without controlling for participants’ musical competence. Evidence suggests that musical training improves pitch-discrimination abilities in speech alongside neural responses to subtle pitch variations in speech (Magne et al., 2006; Moreno et al., 2009). Musicians are expected to demonstrate superior pitch matching skills compared to non-musicians, potentially evidenced by lower duration (τ) and JND (δ) thresholds. Similarly, musical aptitude is expected to correlate positively with pitch-matching performance.

2 Method

2.1 Materials

The voice stimuli used in this experiment were derived from a recording of the vowel [i:] and the syllable [ti:] produced by a male speaker in a sound-treated room. The speaker recorded multiple utterances of the vowel and syllable with level pitch in isolation, using a high-quality Marantz digital audio recorder (PMD661 MK.II) and a Rode Lavalier microphone. The original recording in WAV format has a single channel with a sampling rate of 44.1 kHz and a 16-bit depth.

Baseline vowel token A clear and sonorant token of [i:] in fully modal voice with a relatively flat f_0 track (mean f_0 of 145 Hz) was chosen and resynthesised with stylised baseline f_0 of 150 Hz, using Pitch-Synchronous Overlap and Add (PSOLA) in Praat (Boersma and Weenink, 2024), to create the natural-sounding base token prior to further manipulation of f_0 . A middle portion of the sound (approximately 0.09–0.34s), 250 ms in length with a stable intensity contour, was extracted, of which the boundaries occurred at zero-crossings. The amplitude of the first and last 10 ms of this base token was gradually increased and decreased respectively, using the *Fade in* and *Fade out* commands in Praat (ibid.), to reduce the abruptness of sound onset and termination and prevent auditory artefacts. The baseline vowel was normalised to an average intensity of 75 dB.

Baseline CV token A clear and sonorant token of [ti:] in fully modal voice with a relatively flat f_0 track (mean f_0 of 156 Hz) was selected. This voiceless unaspirated coronal stop has a short-lag VOT of approximately 12 ms. The VOT portion was then cross-spliced onto the beginning of the baseline vowel (150 Hz, 250 ms) to create a CV syllable that differs from the baseline vowel only in the added consonant. A coronal stop was selected because (1) it is widely used across diverse languages, and (2) it mitigates unnaturalness caused by the lack of coarticulation with the cross-spliced high front vowel [i:] (e.g. the absence of velar fronting may sound unnatural).

Baseline pure tone token A pure tone token, with the same length of 250 ms and base frequency of 150 Hz as the baseline vowel token, was created using the *Create Sound as pure tone* command in Praat (ibid.), which also featured a fade-in and a fade-out duration of 10 ms each. The baseline pure tone was also normalised to an average intensity of 75 dB.

For all the baseline tokens, 50 ms silence was padded to both edges to prevent boundary-related information loss resulting from windowing in f_0 -tracking². These zero-padded baseline tokens were subsequently used as input for a Praat script to create tokens with f_0 perturbations.

Stimuli with f_0 perturbations Figure 3 illustrates stimuli for the pitch-matching experiment. Each stimulus was composed of a slope followed by a level tone maintained constant at 150 Hz. The onset frequency varied from 110 Hz to 190 Hz in 10 Hz increments (see Figure 3b); that is, there were four values both below and above 150 Hz ($\Delta f = \pm 10, 20, 30, 40$ Hz, or $\pm 1.12, 2.17, 3.16, 4.09$ semitones). The duration of the slope varied from 40 ms to 120 ms in 20 ms increments ($\Delta t = 40, 60, 80, 100, 120$ ms, see Figure 3a). The overall duration of the stimulus was fixed at 250 ms. The largest temporal extent of the f_0 perturbation approached 50% of the vowel’s

²For instance, when the minimum periodicity frequency is set to be 75 Hz, Praat does not compute f_0 values in the first or last 20 ms of each sound interval because the analysis requires a 40 ms window for every frame.

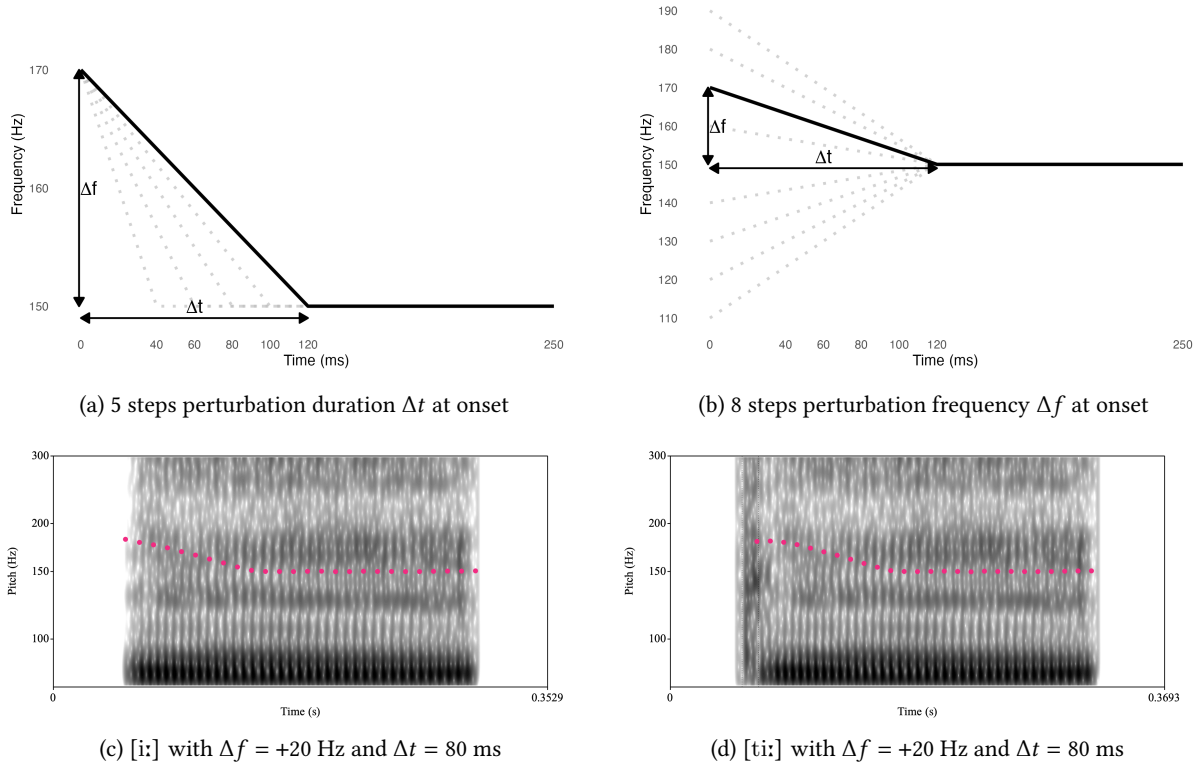


Figure 3: Stimuli illustration. Spectrograms are from 0 to 6000 Hz. The first 10 ms is excluded in Δt given the fade-in effect.

duration, at which point the CF0 effect tends to taper off in many languages. The f_0 manipulation resulted in 8 (perturbation frequency) $\times$ 5 (perturbation duration) = 40 additional distinct f_0 contours. A total of 40 instances of the vowel [i:], 40 instances of the syllable [ti:], and 40 pure tones with varying f_0 contours were resynthesised using the PSOLA algorithm in Praat (ibid.) through a custom script. The periodic intervals of all tokens were normalised to an average intensity of 75 dB.

Response syllable, vowel and pure tone tokens In each condition, response tokens of 250 ms with a steady-state f_0 pattern will also be resynthesised via PSOLA and normalised in the same manner. The f_0 s range from 100 Hz to 200 Hz in 5 Hz increments, resulting in a total of 20 (unique f_0) $\times$ 3 (syllable, vowel and pure tone conditions) = 60 response tokens.

Each target stimulus will be presented 3 times in a randomised order. In total, we will have 45 (40 perturbed tokens + 5 baseline tokens) $\times$ 3 = 135 trials per condition in an experiment. The original recordings and the Praat script are openly accessible to reproduce all the stimuli used in this study.

2.2 Procedure

This study will be programmed using *PsychoPy* (Peirce et al., 2019), a free and versatile software for building experiments in behavioural science, and hosted online via *Pavlovia.org*³ with the open-source JavaScript library, *PsychoJS*. Researchers usually have limited control over the participants' listening environment when conducting online auditory experiments. Requiring the use of headphones (including in-ear and over-the ear versions) is, therefore, essential, as headphones generally provide better sound presentation than speakers.

Considering the limited access to headphones among participants, we will offer two routes of participation: 1) a web-browser-based online experiment; and 2) a complementary in-lab experiment. For online experiments, we will recommend that research labs and participants use Chrome as their browser (Buchanan et al., 2021). Meta-information about the browser and operating system are, nevertheless, saved during the experiment to control for implementation differences. Experiment results of each language will be saved in a separate .csv file, initially available on the Pavlovia server and in the experiment's GitLab repository. All data will then be

³Hosting experiments on *Pavlovia.org* is free with an active institutional licence. There are 150 active site licenses currently. Collaborative teams that have a difficulty in accessing Pavlovia account for free can contact the leadership team for assistance.

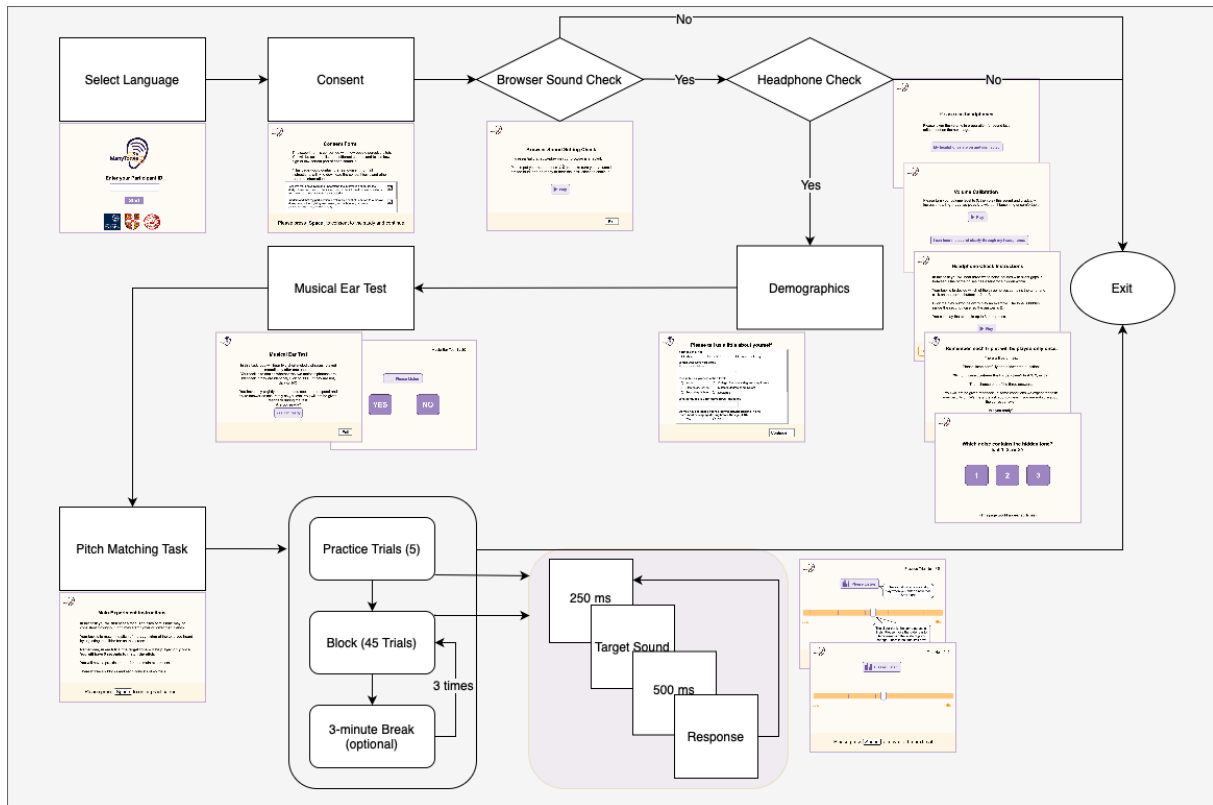


Figure 4: The workflow of the experiment. See supplementary materials for enlarged experimental User-Interface designs.

archived on Github and we will use Zenodo (<https://zenodo.org/>) to release versions of the data with citable DOIs, following Buchanan et al. (2021).

Figure 4 demonstrates the workflow of the experiment, illustrated with intended experimental User-Interface designs (see complete version at our OSF repository). Participants will be required to complete the study using a computer, instead of a mobile or tablet device. This requirement allows for tracking of device’s display, providing meta-data on screen size, browser type, and timing accuracy. Participants will hear the stimuli through headphones at a comfortable high volume level. By pressing the spacebar to consent to participate in this study, participants will first complete a rapid browser sound check and a headphone screening test based on Huggins Pitch (Milne et al., 2020). Upon passing these checks, participants will be asked to complete a brief demographic questionnaire, which collects information on their sex at birth (male, female, prefer not to say), year of birth, educational level (none, elementary school, secondary school, bachelors, masters, doctorate), native (first) language, and previous musical experience. Participants who have at least 6 years of formal private lessons in one instrument or singing starting before the age of 10 will be flagged as the musician group, following the definition of musicianship in Wong and Perrachione (2007). After questionnaire, participants will be prompted to take a brief Musical Ear Test to measure their musical competence.

Then the main experiment will begin. Participants will be assigned to one of three counterbalanced groups which determined the order of experimental blocks and one of the stimuli types (i.e. CV syllables, vowels, or pure tones). They will be asked to adjust the frequency of a flat tone so that it is equal to the frequency at the **beginning** of the target stimuli. The first block of 5 practice trials is an interactive tutorial which oriented participants to the task and controls. Subsequently, stimuli will be presented in three blocks of 45 trials with an optional three-minute break, and trials within each block are pseudo-randomised to prevent systematic biases. Each trial starts with a target stimulus sound, followed by a 500 ms pause and an initial response sound with a level tone of 150 Hz. On the screen, a slider bar ranging from 100 Hz to 200 Hz, with a granularity of 5 Hz, will be displayed and activated. The slider bar is initially positioned at the midpoint of its range (150 Hz), matching the level f_0 part of the target stimulus. The step size of the slider bar corresponds to the differential threshold for a frequency of 140 Hz (Rossi, 1971), assuming that the f_0 glissando threshold is not finer than the differential threshold. If she/he is satisfied with this pair of stimulus and response, the participant confirm their match selection by pressing the spacebar and the next stimulus sound will play after an inter-trial interval of 250 ms. If

she/he is not satisfied with this pair, the participant can adjust the slider bar using a mouse or touchpad to listen to another pair of the stimulus and the selected response. Participants can listen to multiple pairs of sounds before reporting a match. The number of pairs listened for each stimulus will also be recorded. Upon successful completion, participants will either receive payment or course credit for their participation in the study. An estimate for the amount of time required for a participant is approximately 25 to 45 minutes.

2.3 Participants

Participants should be between 18 and 40 years old⁴, although additional age groups can be included in subsequent studies. Additionally, participants should have normal hearing and vision or vision corrected to normal.

Given that previous studies did not explore as many levels of Δf , Δt , and various conditions, nor did they provide accessible data to estimate effect sizes, the estimation of minimal sample size through an a-priori power analysis is not applicable. Nevertheless, a post-hoc power analysis will be conducted using the *simr* package in R (Green and MacLeod, 2016).

In line with traditional practice⁵, we recommend a minimum of 30 participants per group for analysis. There is an inherent trade-off between the number of participants and the length of an experiment session. A within-speaker cross-condition design would require 135 trials per condition across three stimuli conditions, resulting in a total of 405 trials per participant. Considering that fatigue and loss of concentration tend to increase with longer experiments, a more lightweight experiment design—where each participant engages in only one stimuli condition—is preferred, especially for online experiments. This approach, however, necessitates a larger sample size to ensure robust results. For this study, we aim for a gender-balanced group of at least 30×3 (stimuli conditions) = 90 participants per language, but encourage enrolling additional participants to mitigate potential data loss or rejection.

Due to resource limitations, for some participating laboratories and fieldworkers, it may not be feasible to recruit 90 participants. In such cases, priority should be given to recruiting participants for experiments involving the linguistic stimuli (vowel and CV syllable). Laboratories with the capacity and willingness to recruit more than 90 participants are encouraged to include musician groups.

Additionally, laboratories and fieldwork researchers recruiting participants from the same language variety are encouraged to collaborate (e.g., Lab A recruiting musicians while Lab B recruits non-musicians).

3 Scalability Plan

3.1 Language varieties

In order to examine the influence of a native language on the CF0 perception, we plan to include participants from both tonal and non-tonal languages, as well as from languages with different voicing and aspiration contrasts. While many studies have focused on CF0 production, as shown in Table 2 and Table 1, recruiting participants from these languages enables discussions of Δf and Δt with reference to their acoustic data. However, we remain **open for as many languages** to be a part of this project, to create a large-scale, typologically diverse CF0 perception dataset.

3.2 Research collaboration

Research laboratories and fieldwork researchers are invited to participate in this project as collaborators. The leadership team will gather basic information about the participating teams, including their selected language variety, data collection location, data collection mode (i.e. in-lab or at-home experiments), technical and statistical expertise, and funding availability. The leadership team will provide research support while supervising the research progress. The ManyLanguages Canvas platform will be used to connect collaborators and OSF repository to share perception data. This project should result in at least an article for publication and contributors to this project will be co-authors of the publication(s).

⁴According to (Yamasoba et al., 2013) Figure 1, people’s hearing ability does not vary significantly under the age of 40, although this is based on the US population and may vary across languages.

⁵The number of 30 typically allows the Central Limit Theorem to apply, making the sample mean distribution approximately normal.

3.3 Research ethics and funding

Research teams should obtain ethical approval from their respective institutions, or an affiliated institution if their institution lacks an ethical approval process. To support all applications, we will provide a comprehensive package of ethics-related documents.

Funding for conducting the study and for compensating subjects is key to the success of our project. The current leadership team on this project has secured some funding. Interested laboratories and fieldwork researchers that have a difficulty with funding can contact the leadership team for assistance. We are willing to help collaborators pursue various national and international avenues for funding. As a principle, collaborating laboratories that already have funding, could be paired with laboratories without funding, so as to enlarge the ambit of languages that inform our results.

3.4 Spin-off projects

We envision expanding this project (under the macroproject name *ManyTones*⁶), where a series of potential spin-off projects could build on this initial experiment, following a similar online workflow. For instance, participating laboratories can further explore the influence of demographic variables, such as musical talent (e.g., possession of absolute pitch) and age (e.g., children versus adults, young adults versus older adults), on pitch discrimination sensitivity. Furthermore, investigating the impact of baseline pitch height, contour shape, position of perturbation (onset versus offset), and varying noise levels on participants' performance could yield significant insights. In addition, previous investigations have suggested that the observed participants' sensitivity to pitch is highly task-relevant. Participating laboratories can also compare the results of the main project with findings from similar studies using other types of task (e.g., 2AFC task), to explore potential methodological differences. These investigations have the potential to deepen our understanding of the underlying mechanisms and expand the scope of the original study.

4 Analysis Plan

4.1 Data Structure and Variables

Table 3 outlines the comprehensive data structure for our study, detailing each variable collected and its characteristics. Each variable will be a column in the data sheet.

The "Participant ID" column, which is categorically coded, records participants' native language background using a three-digit language code and assigns a unique identifier within each language group. The "Block" column categorises the block numbers, tracking the sequential progression of the experiment. "Trial Number", a numerical variable, records the sequence of stimulus presentations, aiding in the identification of potential fatigue or practice effects. The "Item ID", also categorical, indicate specific stimuli items.

The categorical "Condition" column differentiates between vowel, pure tone, and consonant-vowel (CV) syllable stimuli. Musical experience is also defined categorically (Yes/No). The Musical Ear Test (MET) score is a numerical value indicating a participant's accuracy on the MET. Both musical experience and the MET score are included in the analysis for three reasons. First, while musical experience provides insight into whether formal musical training affects pitch-matching capacity, the MET score measures innate musical ability—it is possible for individuals with perfect musical sense to have never received formal training. Second, musical experience varies cross-culturally, and no consensus exists on the duration or type of musical experience necessary for meaningful comparison. Third, for language varieties with limited participant availability, it may be challenging to recruit sufficient numbers for both musical and non-musical groups. In such cases, the numerical MET score can serve as a regression indicator of musical ability, offering a standardized alternative to musical experience.

Perturbation frequency (Δf) and perturbation duration (Δt) are categorised with multiple values to capture variations in the stimuli. The perceived direction of the perturbation will be recorded in "Response Δf ", using positive or negative values. "Directional Accuracy" is a binary categorical variable indicating whether the response direction was correct. The "Onset frequency" and "Response Frequency" are numerical measures, and these can be used to calculate an "Error Distance". The "Error distance" measures the absolute difference between the response frequency and the onset frequency, providing a numerical error metric.

Reaction time (RT) and its log-transformed counterpart (logRT) are recorded. The main purpose of including the RT data is to filter out potentially invalid responses. Finally, "Demographics" columns allow for the inclusion

⁶See logo design for ManyTones in our OSF. A dedicated website for hosting all relevant projects is under construction.

Column Name	Example	(Approximate) Number of values	Description
Language Code	ENG	NA	Categorical
Participant ID	ENG101	Min. 90 per language	Categorical; Language Code + Participant Number
Group	A	3	Categorical
Block	1	3	Categorical
Trial Number	1	135	Numerical
Item ID	v-30p-100	45*3 (Condition)	Categorical
Condition	Vowel	2	Categorical; 3 levels: vowel, pure tone, CV syllable
Music Experience	Yes	Min. 90*135	Categorical
Music Ear Test Score	60%	90*54	Numerical
Δf	10Hz	9	Categorical; 4 positive values, 0, and 4 negative values
Δt	40ms	5	Categorical
Onset Frequency	190Hz	9	Numerical; only recorded in the onset task
Number of match attempts	5	NA	Numerical (integer)
Response Δf	15Hz	Min. 90*135	Numerical
Response Frequency	180Hz	Min. 90*135	Numerical
Directional Accuracy	Correct	Min. 90*135	Categorical (binary)
Error Distance	5Hz	Min. 90*135	Numerical; the absolute difference between the response frequency and the onset frequency
Reaction Time (RT)	2000ms	Min. 90*135	Numerical
Log-Transformed RT (logRT)	3.5	Min. 90*135	Numerical
Demographics	NA	NA	Can be any relevant demographic information, e.g., age, sex, education, etc.
Experiment Setting	Web-based	2	Categorical; Web-based or In-person
Device Information	NA	NA	Can be any relevant device information, e.g., screen size, browser, IP address, etc.
Language Information	NA	NA	Can be any relevant language information, e.g. family, tonal status, number of tones, voicing contrast, stress pattern, etc.

Table 3: Sample Data Structure. Note that “Demographics”, “Device Information”, and “Language Information” are place-holders for demographic-related and device-related variables respectively.

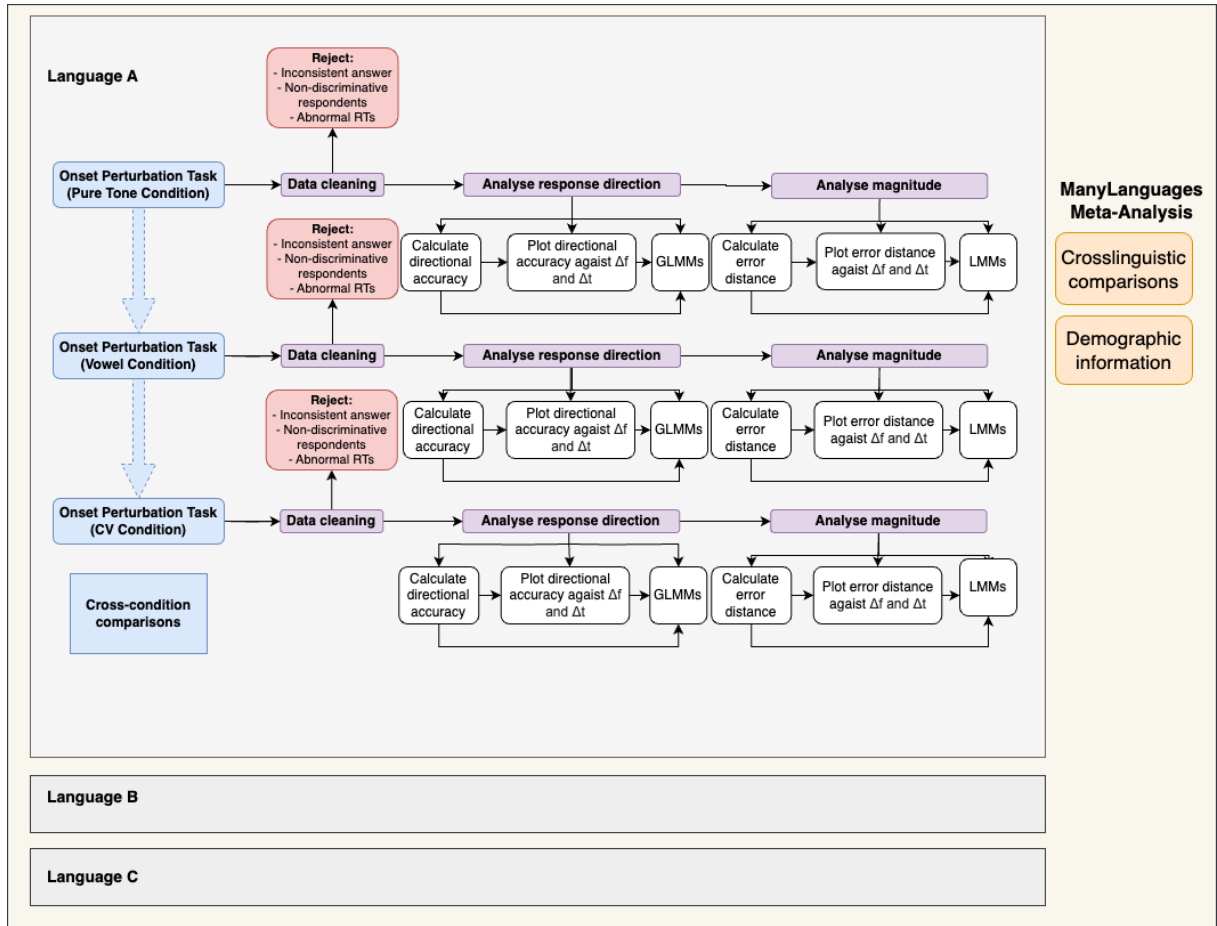


Figure 5: Data analysis procedure

of relevant participant information such as age and gender, facilitating a more detailed analysis of the data. The “Display Information” columns will record participants’ device information, such as the type of computer/laptop, screen size, IP address. This information may be used to account for within-group or within-condition variance.

4.2 Statistical Analysis Procedure

The data analysis for the project is structured into three key steps: (1) data pre-processing; (2) analysis of the perception of f_0 perturbation direction; and (3) analysis of the perception of f_0 perturbation magnitude. This analysis will be exemplified in the following subsections using data from one language under one condition. We will focus on analysis within each language variety before embarking on crosslinguistic comparisons. For the whole analysis procedure, please refer to Figure 5.

4.2.1 Data pre-processing

The first step is data pre-processing and cleaning. We will exclude participants who consistently move the slider bar in the same manner regardless of the stimulus, those who exhibit highly inconsistent responses across different repetitions of the same stimulus, and those whose overall response times are either too quick or too slow (mean reaction time outside the range of mean ± 2.5 standard deviations). Additionally, trials affected by computer errors or missing data (i.e., participant inattentiveness and timeout trials, internet disconnection, computer crashes) will be marked as ‘NA’ in the final dataset.

4.2.2 Perception of the directionality of perturbations

In the next step, we examine how accurately participants can perceive the direction of pitch change (i.e., whether the perturbation is higher or lower than the base pitch). By analysing how directional accuracy changes as the perturbation increases, we can address the second objective of this project (see Section 1). This provides detailed

information on the specific Δf level (in Hz) at which participants' perception of pitch change direction becomes highly accurate and whether this change is gradual or abrupt. Specifically, the directional accuracy is defined as the proportion of correct directional responses out of the total responses. If the pitch change direction in the response matches the perturbation direction in the target stimulus, the directionality is coded as “correct”; otherwise, it is coded as “incorrect”. To further investigate how participants' perception of direction changes with Δt and Δf , a Generalised Linear Mixed-Effects Model (GLMM) will be built with the directional accuracy as the predicted variable, Δt , Δf , and their interactions as fixed effects, and participant and item as random intercepts, using the lme4 (Bates et al., 2015) package in R (R Core Team, 2021). Before building the model, variables that are not of primary interest, such as sex, age, musical experience, MET score, and block number, will be explored and compared against the null model. This step aims to identify whether these variables have any significant effect on accuracy and to determine if they need to be rigorously controlled in the analysis. Before finalising the best-fit model, assumptions will be checked using the DHARMA package (Hartig and Hartig, 2022).

4.2.3 Perception of the magnitude of perturbations

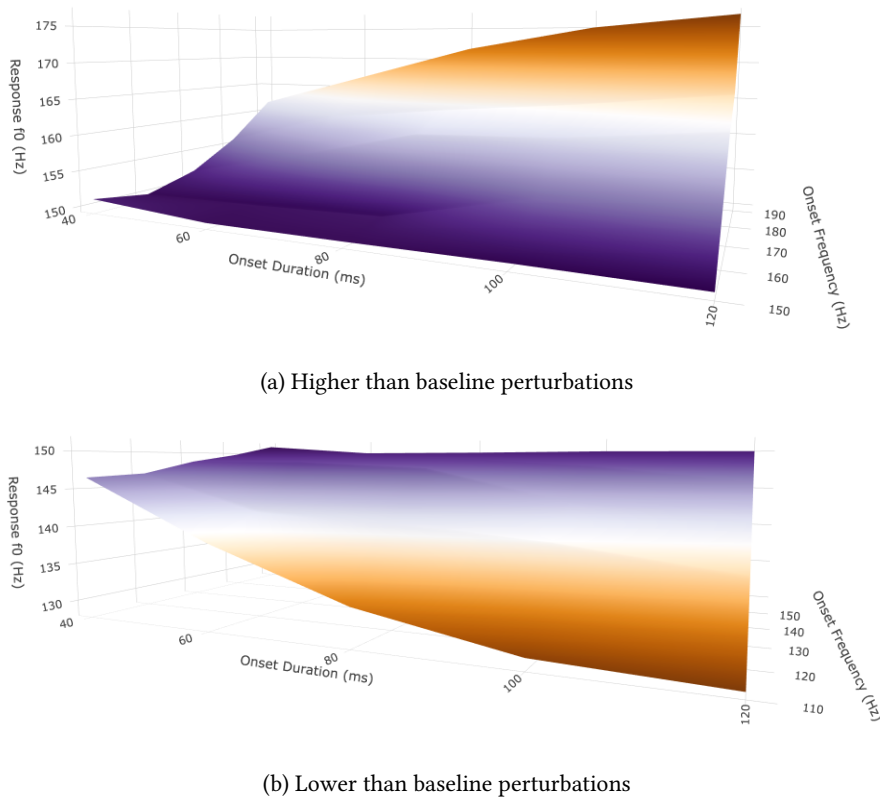


Figure 6: Demonstration of participants' responses in simulated data.

For the third step, we investigate how accurately participants quantify changes in pitch across an array of perturbation scenarios (Δf and Δt configurations). The average response frequency can be visualised as in Figure 6. We can further calculate the distance between the response onset frequency and the onset frequency of the stimulus (termed “error distance”) and visualise the error distance against Δf and Δt in a three-dimension plot for each condition. Then, a Linear Mixed-Effects Model will be built with the error distance as the predicted variable, the Δf , Δt , and their interactions as predictors, with participant and item as random intercepts. Before building the model, variables that are not of primary interest—such as gender, age, musical experience, MET score, and block number—will be explored and compared against the null model. This step aims to identify whether these variables have any significant effect on accuracy and to determine if they need to be rigorously controlled in the analysis. Before finalising the best-fit model, assumptions will be checked.

To explore whether the condition of the stimuli (i.e., pure tone, vowel, or CV syllable condition) will have an effect, we then conduct between-condition comparisons. Two mixed-effects models will be built, treating the directional accuracy and error distance as the predicted variables respectively, Δf , Δt , condition, and their interactions as predictors, and participant and item as random intercepts.

4.2.4 Cross-linguistic analysis

The within-variety analysis procedure mentioned above can be conducted by individual laboratories. An alternative option for individual fieldwork experimenters is to collaborate directly with the leadership team or another participating laboratories who have the expertise in data processing and statistics to perform the analyses.

Once results are obtained from multiple languages across research teams and labs, we can perform a cross-linguistic meta-analysis. This involves observing the cross-linguistic data visually to identify potential factors causing the differences. We then create plots and build models based on these observations. The ultimate goal is to provide a comprehensive meta-analysis and a systematic account for the effects on onset f_0 perturbation perception.

5 Supplementary Materials

The supplementary materials are available at our OSF repository: <https://osf.io/ykte6/>.

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